

D VARSHITH REDDY

AIML Student — Full Stack & AI Engineer

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Technical Skills

Programming Languages	Python, Java, JavaScript, C, HTML, CSS
Web Development	React.js, Next.js, Tailwind CSS, Node.js, Express.js, FastAPI
Databases	MongoDB, PostgreSQL, Firebase Firestore
AI/ML & Tools	TensorFlow, OpenCV, Machine Learning, AI Integration, Fraud Detection
Platforms & Services	Git, GitHub, Docker, AWS, Mapbox API, Leaflet.js, OpenStreetMap

Education

Maturi Venkata Subba Rao Engineering College (MVSR)

2028

B.E. in CSE — Artificial Intelligence & Machine Learning

Hyderabad, India

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning, Artificial Intelligence.

Projects

KAVACH AI — Sovereign Agentic AI Workbench

- Developed a self-hosted, air-gapped agentic AI workbench for confidential industrial knowledge work using open-weight multimodal LLMs and local inference.
- Designed multi-agent routing across local models, tools, and knowledge bases with Retrieval-Augmented Generation (RAG), multimodal P&ID analysis, and evidence-backed responses.
- Built an auditable workflow with live agent traces, source verification, security monitoring, human-in-the-loop approval, and PDF, DOCX, and XLSX deliverable generation.

Collections360 — Intelligent Collections Decision Platform

- Built a unified **Collections 360** data platform integrating CRM, cards, loans, deposits, and bureau data into a governed Golden C360 customer record with lineage.
- Developed grounded Q&A and analytics workflows using **NL-to-SQL**, certified metrics, SQL guardrails, policy retrieval, and source-backed responses for collection agents.
- Implemented an uplift-based Next Best Action model with **SHAP explanations** and hard policy gates for safe, explainable, and human-approved collection recommendations.

TraceNet AI — Geo-Intelligent Lost & Found Recovery Platform

- Built a geo-intelligent lost & found platform with interactive maps, radius-based tracking, and real-time location visualization using **Leaflet.js** and **OpenStreetMap**.
- Developed AI-based smart matching and image similarity analysis with confidence scoring, fraud detection, and **TrustScore** verification.
- Designed analytics dashboards with hotspot visualization, duplicate filtering, recovery tracking, and secure coordination using **Firebase**.

HelioMind AI — AI-Powered Rooftop Solar Intelligence Platform

- Built an AI-powered rooftop solar intelligence platform that analyzes roof imagery, scores solar suitability, parses electricity bills, and recommends pay-per-watt installations.
- Developed computer vision pipelines using **YOLOv8** and **OpenCV** to detect roof segments, obstructions, and usable installation areas for automated rooftop analysis.
- Implemented electricity bill OCR, weather-aware irradiance and seasonal yield projections, solar suitability scoring, and transparent pay-per-watt cost estimation.
- Built an interactive dashboard with savings, ROI, carbon-offset analytics, and an AI assistant using **Next.js**, **FastAPI**, **PostgreSQL**, **Docker**, and cloud deployment infrastructure.

Leadership & Activities

- Currently working as a **Research Student** at MVSR Engineering College on a research project focused on **Battery Management Systems (BMS)**, involving intelligent monitoring, optimization, and energy efficiency analysis.
- **Secured a Top 10 position at the IIIT Hyderabad Infinium Hackathon 2026**, competing among teams in an AI-focused, national-level hackathon.
- Participated in **10+ hackathons** as a **Team Lead**, collaborating on AI-integrated software solutions, rapid prototyping, and real-world problem solving.
- Core Member of the college **AIML Club**, contributing to AI, ML, and software engineering activities.
- Machine Learning Student Representative, supporting AI/ML initiatives, student engagement, and technical activities within the department.